
Tensile Fracture and Bond Properties of Ductile Hybrid FRP Reinforcing Bars

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SUMMARY

Fibre reinforced plastic (FRP) rebar has only linear elastic behaviour, whereas steel rebar has linear elastic behaviour up to its yield point, followed by a large amount of plastic deformation. With no plastic deformation, a small increase in the load acting on concrete structures containing FRP rebar can cause a catastrophic collapse without warning. A ductile hybrid FRP rebar was manufactured and evaluated in this study. The tensile and bond strengths of the hybrid FRP rebar were tested to determine its fracture properties and bond behaviour. It showed elastic behaviour up to the point of early fracture, but had very irregular behaviour thereafter. The fibres in the sleeves were broken at very irregular lengths, while the cores appeared to fracture in a regular pattern. The stress-strain curves were linearly elastic, with a definite yield point followed by pseudo-plastic deformation. In addition, the hybrid rebar specimens exhibited a large amount of slip when the peak load was reached.

1. INTRODUCTION

The deterioration of the structural performance of reinforced concrete structures as a result of the corrosion of steel reinforcement is a major concern¹⁻⁶. Such reinforced concrete structures are used mainly in infrastructure, and the problem is much greater in concrete that is in direct contact with water, such as marine structures and concrete bridges. Although appropriate corrective measures are required, a specific solution has not yet been proposed. In the United States, 180,000 out of 580,000 bridges, or 31%, require immediate repair or reinforcement owing to corrosion defects⁷⁻⁹. Among the alternatives developed to solve the corrosion problem, FRP reinforcements show great promise as an efficient solution^{1-3,7,8,10,11}. They have high strength and present no risk of corrosion^{1-3,8,10-12}. They can reduce the dead load of the structure with their high strength to weight ratio^{1-3,5,13}. However, they are brittle and they yield without undergoing any prior plastic deformation^{7,8,11}. In

other words, they have linear elastic stress-strain characteristics.

By contrast, steel reinforcements have linear elastic characteristics until their initial yield point, which is then followed by a large amount of plastic deformation and by strain-hardening characteristics^{7,8,11}. The plastic deformation continuously distributes the load and, at the same time, produces a large amount of ductile bending until the member breaks. Without plastic deformation of the steel reinforcements within concrete structures, even a slight increase in the breaking load could result in sudden brittle failure. In addition, plastic deformation is necessary for the design of concrete structures that are subjected to dynamic loads by earthquakes, wind, or vehicles, because the plastic behaviour distributes the energy resulting from the deformation of the material, and absorbs much of the energy applied to the structure by repeated loads^{7,8,11}.

In this study, hybrid FRP rebar was manufactured to overcome the brittleness problems of existing commercial FRP rebar based on a design method developed previously⁸. The tensile behaviour, fracture properties, and bond performance of the hybrid FRP rebar were also evaluated.

2. MANUFACTURE OF HYBRID FRP REBAR

In this research, three types of hybrid FRP rebar were manufactured, based on a design method suggested by Somboonsong *et al.*⁸ The target performance of the hybrid FRP rebar was as follows:

- Hybrid FRP rebar (A): tensile modulus = 60 GPa, strain $\geq 3.0\%$
- Hybrid FRP rebar (B): tensile modulus = 40 GPa, strain $\geq 3.0\%$
- Hybrid FRP rebar (C): tensile modulus = 35 GPa, strain $\geq 3.0\%$

In addition, to design hybrid FRP rebar that meets the target performance, we determined the mixture ratio of the fibre, twisting angle, crimp angle, and braiding angle based on a previous design method⁸. **Table 1** lists the design values applied in this study.

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Table 1. Design values applied to the manufacture of hybrid FRP rebar

Design parameter	Materials		Mix ratio (volume ratio)			Twist angle		Crimp angle		Braiding angle	
	Core	Sleeve	Carbon	Aramid	Glass	Aramid 10tex ^{1/2} ×1/cm	Glass 10tex ^{1/2} ×1/cm	Aramid	Glass	Aramid	Glass
Carbon + Aramid	Carbon	Aramid	27	73	-			10	-	25	25
Carbon + Glass	Carbon	Glass	19	-	81			-	30	25	25
Carbon + Aramid + Glass	Carbon	Aramid Glass	13.6	34.6	51.8			10	30	25	25

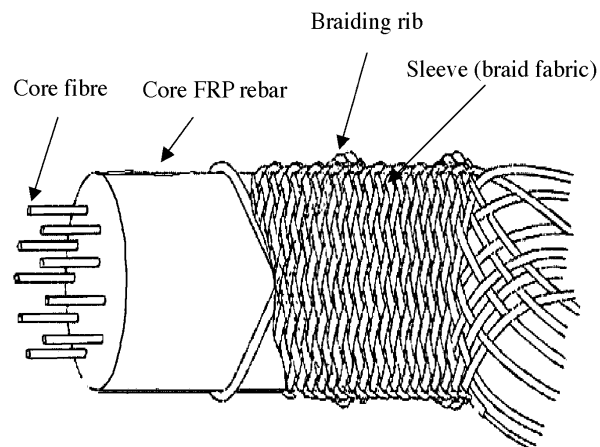
The most important properties of FRP reinforcements are their strength, stiffness, and dimensional stability. FRP reinforcements can be used for concrete structures based on appropriate designs that satisfy all the necessary requirements. For example, glass fibre is very weak in alkali, water, or chemical environments, but it has good toughness compare to carbon fibre. Carbon fibre has very good strength, stiffness, and chemical resistance, but it has a very small impact resistance owing to its brittleness. Aramid fibres are not stronger than carbon fibres, but they are stronger than glass fibres. After considering these characteristics, carbon, aramid, and glass fibres, which are the most widely used in the production of FRP reinforcements today, were chosen as possible working materials in this research. The selected polymer matrix was vinylester resin, which, based on the results of previous research, has good durability against alkali and other outside environments¹⁰. The raw materials used for specimens were carbon fibre (amoco T-300), aramid fibre (technora), E-glass fibre, (hankuk fibreglass ECK 18) and vinyl ester resin (Ashland chemical D-1618). The characteristics of the fibres and the matrix resin used in this research are shown in **Table 2**. Also, the vinylester resin cured at 150°.

The advantage of a core-sleeve structure is that it is easy to reach the target performance of rebar by controlling the geometric variation of the fibres that form the core and sleeve⁸.

The fibre used in the core of the hybrid FRP reinforcements affects the overall elastic modulus, whereas the fibre in the sleeve affects the ultimate strain. Therefore the elastic modulus of the FRP reinforcements increases with the elastic modulus of the core fibre. In addition, the ultimate strain of the reinforcements decreases as the elastic modulus of the sleeve fibre increased. Considering this relationship and previous research results⁸, we selected carbon fibre as the material for the core of the FRP reinforcements. **Figure 1** shows the core and sleeve structure of the hybrid FRP rebar.

After the reinforcement core fails, the applied load must be supported by the sleeve fibre. We considered both aramid and glass fibres as possible sleeve materials for a FRP reinforcement made from a carbon–aramid–glass mixture. In addition, we believe that the carbon and glass fibre mixture examined in this study not only enhances tensile behaviour, it is also beneficial economically.

Twists, crimps, and braids have the same influence on the core and sleeve fibres of FRP reinforcements. As the amount of twisting, crimping, and

Figure 1. Schematic of the composition of hybrid FRP rebar**Table 2. Mechanical properties of the fibres and matrix**

Mechanical properties	Matrix (Vinylester)	Fibre type		
		E-glass	Aramid	Carbon
Yield stress (MPa)	90	1,890	3,100	3,500
Elastic modulus (GPa)	3.4	75	77	244
Ultimate strain (%)	5.2	2.64	4.6	1.21
Fibre density (g/cm ³)	-	2.62	1.39	2
Fibre diameter (10 ⁻⁶ m)	-	16.5	12	10

braiding increases, the initial elastic moduli of the core and sleeve fibres decrease and the strains after yielding increase⁸.

Therefore, we designed the core of the FRP reinforcements by linearly arranging fibres without any twists. However, as the amount of sleeve fibre twisting increased, the elastic modulus decreased, while the ultimate strain increased. It was very difficult to design a fibre without any twists because of the characteristics of the production system, except when the fibre was extruded as set. In particular, when braiding equipment was used to produce the sleeve fibres for the FRP reinforcements, it was almost impossible to prevent the fibres from twisting because the fibre was woven while it was being supplied to the three-dimensional system. Therefore, we selected 0.1 twists/cm for the sleeve fibre twist, T . (The units of T can be expressed as the number of fibre twists per centimetre; $10 \text{ tex}^{1/2} \times 1/\text{cm}$ indicates 0.1 twists/cm). This is the minimum value for our production equipment, based on the design method used in previous research⁸. As the crimp angle increased, the elastic modulus of the reinforcements decreased and the ultimate-strain-to-yield-strain ratio increased⁸. Therefore, the minimum tolerance was used for the crimp angle. It was determined during the process of braiding the sleeve fibre. After considering the crimp angle range of the production system along with the elastic modulus and pseudo-plastic deformation (3%) actually required for the reinforcement, we selected a crimp angle of 10° for the aramid fibres and 30° for the glass fibres, based on the previous design method⁸.

The elastic modulus of the hybrid FRP reinforcement decreased and the ultimate-strain-to-yield-strain ratio increased as the braiding angle of the sleeve fibre increased⁸. The value of the braiding angle involved a certain tolerance because of the three-dimensional manufacturing system.

Considering both the characteristics of the manufacturing system and the previous results⁸, we selected 25° for the braiding angle for both the aramid and glass fibres; this value maximises the elastic modulus, while maintaining the desired 3% pseudo-plastic strain.

The modulus of elasticity and the strain of three kinds of hybrid FRP rebar derived from applied design values in **Table 1** and the materials properties in **Table 2** are as follows:

- Hybrid FRP rebar (A): tensile modulus=63 GPa, strain=3.4%
- Hybrid FRP rebar (B): tensile modulus=43 GPa, strain=3.4%
- Hybrid FRP rebar (C): tensile modulus=37 GPa, strain=3.4%

A key parameter that affects the performance of concrete structures is the bond strength between the FRP rebar and concrete^{2,5,12,14,15}. The bond performance depends on the properties of the FRP rebar surface pattern. The surface characteristics of the FRP rebar are an important property for its mechanical bonding with concrete. Three types of surface deformation patterns for FRP rebar that are commercially available include helical wrapping and sand coating, sand coating, and rib moulding². The helical wrapping method cannot provide enough surface deformation

for bonding because helical wrapping fibre and sand coated are easily sheared from the FRP rebar. Sand coating and rib moulding provide surface deformation only to the outer FRP skin and surface sand coating and surface rib can easily delaminate from FRP rebars⁷. To overcome the bond performance problems of FRP rebar, we applied a braided ribbed bar⁷; this is difficult to shear and delaminate from FRP rebar. Moreover, its bond performance is dependent on the width and height of the rib. In this study, the width and height of the rib were 2.0 mm and 1.3 mm, respectively. **Figure 2** shows the geometric surface pattern of the hybrid FRP rebar.

3. TENSILE BEHAVIOUR

Three types of hybrid FRP reinforcement were manufactured (**Figure 3**). They were tested in accordance with ACI 440 K standards and the results were evaluated to determine tensile behaviour. The tensile tests were performed in a 250-kN universal testing machine (ESH Testing Limited, model No. esh 250) with a displacement rate of 5 mm per minute in the displacement-controlled mode. The tensile test results for the hybrid FRP rebar were compared with those for the glass fibre reinforced plastic (GFRP) rebar. **Figure 4** shows the tensile behaviour of the three types of hybrid FRP rebar

Figure 2. Geometric surface pattern of the hybrid FRP rebar

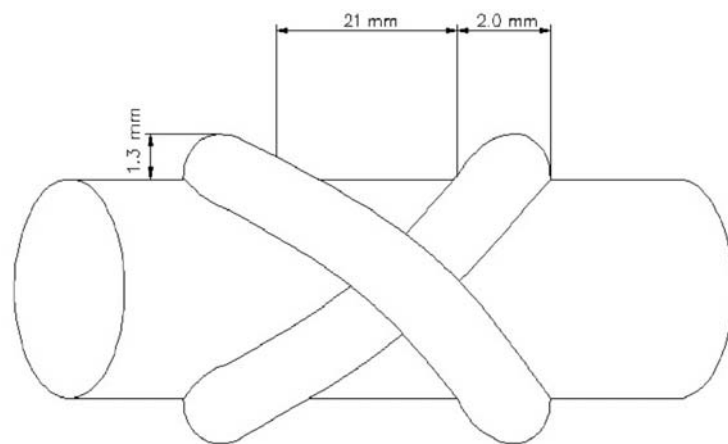
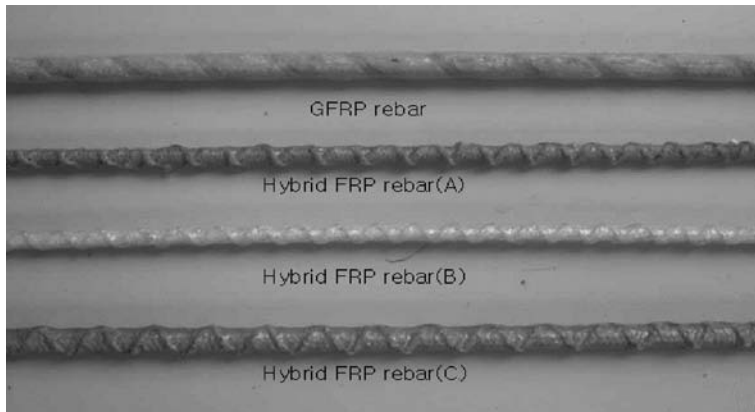
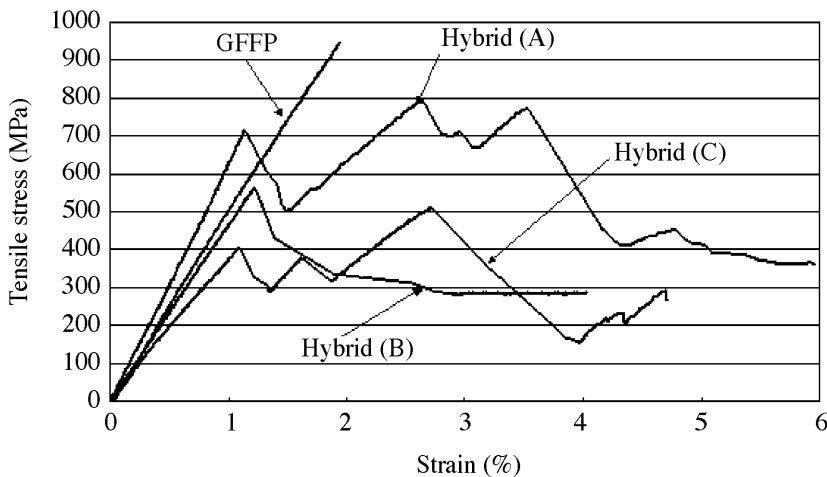


Figure 3. Photographs of the hybrid and GFRP rebar**Figure 4. Tensile behaviour of the ductile hybrid FRP rebar**

and GFRPrebar. GFRPrebar has linear elastic behaviour up to tensile failure, while the hybrid FRP rebar has linear elastic behaviour up to its yield point, followed by a large pseudoplastic deformation.

Although carbon, aramid, and glass fibres are linear elastic, the strain in each is different. In this research, the fibre that had high strain would fail after the fibre that had the lowest strain, as a result of mixing the textiles. The proportions in which the fibres were mixed were based on previous results⁸. In addition, we obtained the target strain as a result of modifying the twist angle, crimp angle, and braiding

angle because the strain in the fibres did not exceed 3%.

For the carbon and aramid fibre rebar (hybrid FRP rebar (A)), the strength of the reinforcements increased and then decreased as the fibre subjected to the greatest strain gradually fractured. Moreover, the carbon and aramid fibre rebar had pseudoplastic deformations exceeding 3%.

For the hybrid FRP reinforcements made of carbon and glass fibres (hybrid FRP rebar (B)), the initial strength drops were very large. Therefore the pseudoplastic deformation seen

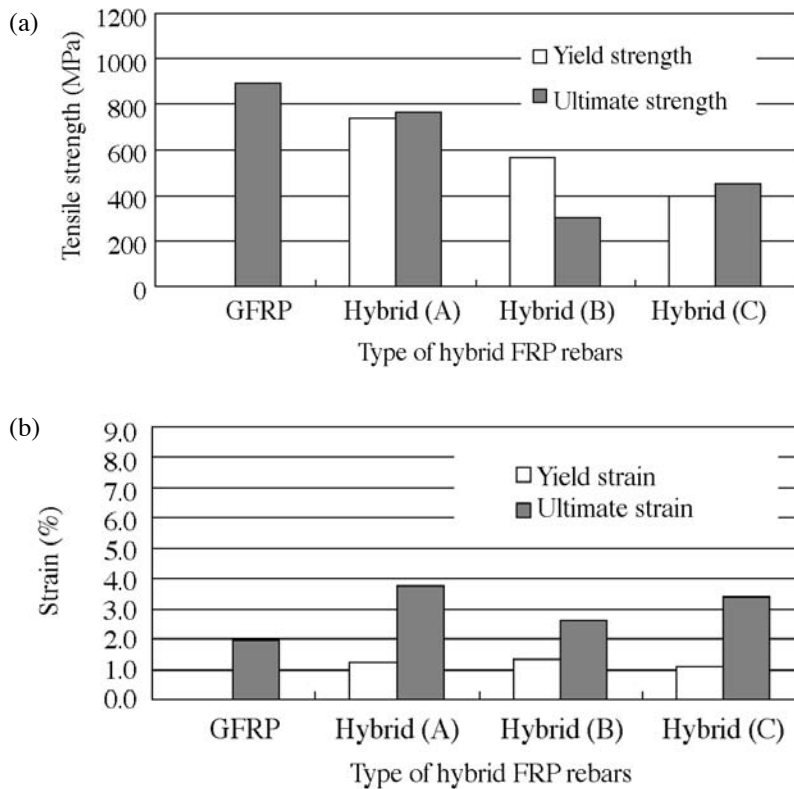
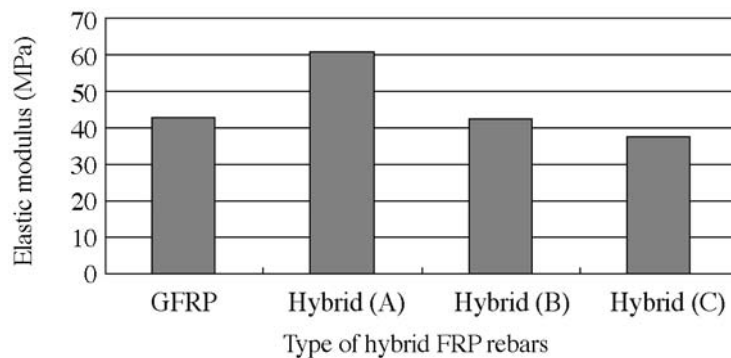
after the initial strength drop shows that the strain was less than 3%. The strength of the rebar increased and then decreased as the deformed rebar gradually fractured.

For hybrid FRP reinforcements with cores made of carbon fibre and sleeves made of glass and aramid fibres (hybrid FRP rebar (C)), the strength of the reinforcements increased and then decreased as the fibres subjected to the greatest strain gradually fractured. In addition, the rebar showed pseudo-plastic deformation exceeding 3%. However, it was not as strong as the carbon and aramid fibre rebar.

Figures 5 and 6 show the tensile strength, strain, and elastic modulus of the three types of hybrid FRP rebar and GFRP rebar. The test results are compared with the target and design values based on the elastic modulus and tensile strain. The test results determined that the elastic moduli of hybrid FRP rebars (A), (B), and (C) were 61, 42 and 37 GPa, respectively. These values were within the target values. Nevertheless, they were lower than the designed values for hybrid FRP rebar (A) and (B) because of the error produced by the twist angle, crimp angle, and braiding angle. However, there was no significant difference between the design values and the experimental results.

The tensile strain of hybrid FRPrebar (A) and (C) was more than 3.4% at fracture, which met the desired target and design values. By contrast, the tensile strain of (B) was low compared to the target and design values. The strain increased when aramid fibres were used as the sleeve, while the strain decreased when glass fibres were used. This resulted from the material characteristics of aramid and glass fibres; aramid fibres fail as a result of necking, while glass fibres undergo brittle failure.

Figure 7 shows the tensile failure mode of the three types of hybrid FRP rebar and GFRP rebar. From **Figure 7**, the

Figure 5. Tensile test results for the hybrid FRP rebar: (a) tensile strength, (b) strain**Figure 6. Elastic modulus of the hybrid FRP rebar**

tensile failure mode of hybrid FRP rebar is necking failure, while the GFRP rebar undergoes brittle failure.

4. FRACTURE PROPERTIES

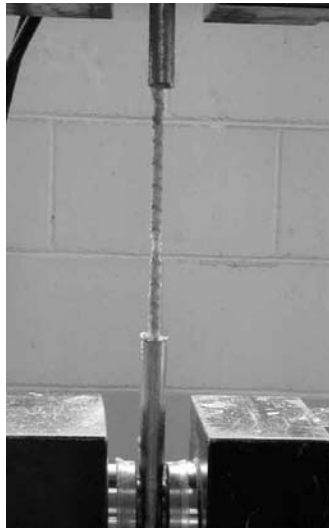
The tensile test results showed that hybrid FRP rebar undergoes pseudo-plastic deformation. However, the fracture properties differed slightly because the strength of the hybrid FRP rebar reinforcements continued

to increase and then decrease after the yield point was reached. **Figures 8–10** show cross sections of the hybrid FRP rebar after fracture; the figures show that the fibres in the sleeves of the hybrid FRP rebar (A), (B), and (C) broke at very irregular lengths, while the cores appeared to fracture in a regular pattern.

In **Figure 8**, the failure pattern of the fibres is very irregular for the aramid

Figure 7. Tensile failure modes of the hybrid FRP rebar**(a) GFRP****(b) Hybrid (A)****(c) Hybrid (B)**

Figure 7. Cont'd...



(d) Hybrid (C)

fibres forming the sleeve because of the material characteristics of aramid fibres, which undergo failure by necking. We observed an increase in continuous plastic deformation after the initial yield in the stress–strain curve. **Figure 9** shows that the glass fibres did not fail as a result of necking, but the glass fibres underwent brittle failure after a small extension in the crimped and braided parts because the glass fibres that form the sleeve are brittle. Therefore the glass fibres had a smoother failure pattern than the aramid fibres shown in **Figure 8**, and no large increase in plastic deformation appeared after the initial yield in the stress–strain curve. In **Figure 10**, the glass fibres show brittle failure, while the aramid fibre show necking failure at the same time

Based on these results, the hybrid FRP rebar showed elastic behaviour up to the point of early fracture, while it underwent very irregular behaviour after the early fracture. Therefore, it was relatively easy to determine the early yield strengths of the FRP rebar, but it was more difficult to determine the elastic modulus and the behaviour of the sleeves correctly.

Figure 8. Fracture surface of hybrid FRP rebar (A): (a) side view, (b) plane view

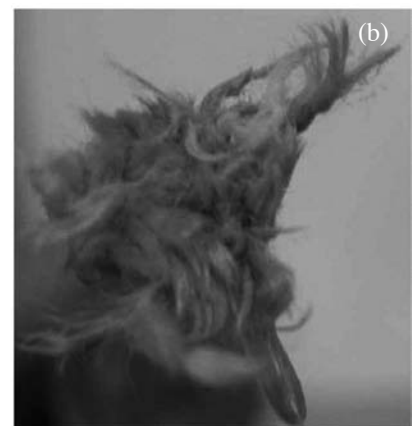


Figure 9. Fracture surface of hybrid FRP rebar (B): (a) side view, (b) plane view

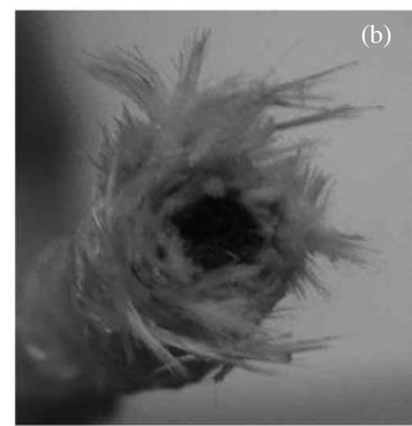
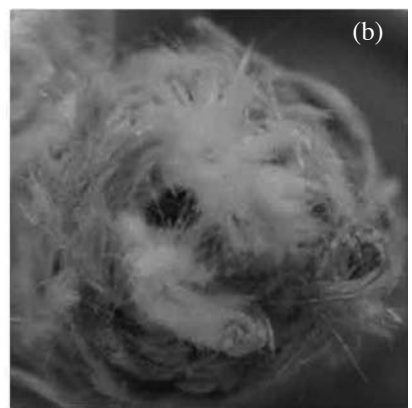
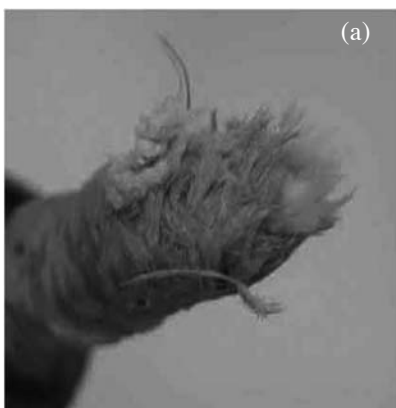


Figure 10. Fracture surface of hybrid FRP rebar (C): (a) side view, (b) plane view



5. BOND BEHAVIOUR

The bond test method used in this study was the direct pullout test method. The pullout specimen consisted of a concrete cylinder measuring Φ

150×300 mm with a hybrid FRP rebar placed concentrically through the end of the cylinder to allow slip measurements¹⁵. The embedment length was adjusted to five times the diameter of the rebar. The mean

compressive strength of the concrete used throughout this study was 36 MPa at 28 days of age. The bond test was performed with a machine load capacity of 25 kN and the rate of crosshead motion was 5 mm/min. The following equation was used to calculate the bond strength:

$$\tau = \frac{P_{max}}{2\pi rL} \quad (1)$$

where τ = bond strength, P_{max} = maximum bond load, r = diameter, and L = embedded length.

The bonding strength-slip behaviour is shown in **Figure 11**. The hybrid FRP rebar specimens exhibited a large amount of slip when the peak load was reached. They continued to hold a relatively high load and to slip before the load was finally reduced. By contrast, the GFRPrebar specimens exhibited only a small amount of slip when the peak load was reached; then, as the slip continued, small peaks in their bonding stresses occurred. The bond strengths are shown in **Figure 12**. According to the bonding test results, the hybrid FRP rebar specimens had greater bonding strength with concrete than the GFRP rebar specimens.

6. CONCLUSIONS

This study involved the manufacture and evaluation of the tensile behaviour, fracture properties, and bond performance of three types of hybrid FRP rebar. The following conclusions were drawn.

- Based on previous results, the hybrid FRP reinforcements manufactured in this study performed very well, with pseudo-plastic deformation of 3% or more. This should ensure the safety and reliability of those structures subjected to dynamic loads by earthquakes, wind, or vehicles. This cannot be achieved using existing FRP reinforcements.

Figure 11. Bond strength-slip behaviour of the hybrid FRP rebar

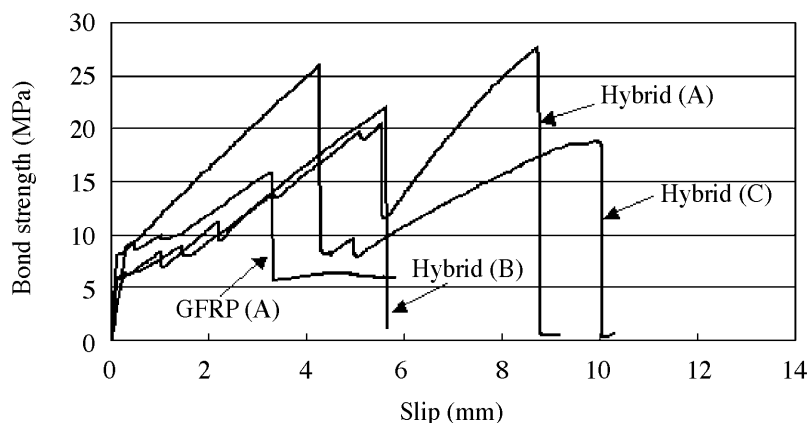
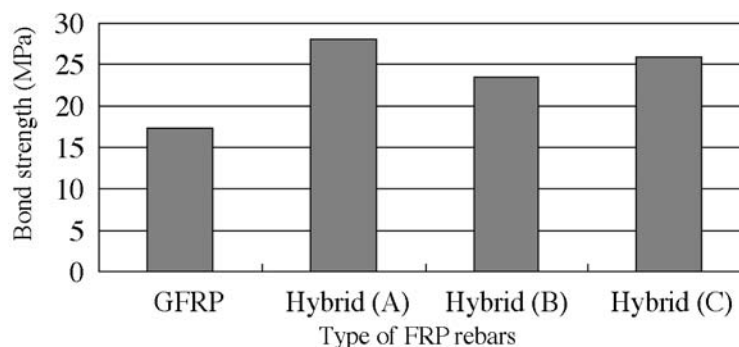


Figure 12. Bond strength of the hybrid FRP rebar



- The hybrid FRP rebar showed elastic behaviour up to the point of early fracture, but had very irregular behaviour thereafter, since the fibres in the sleeves of the hybrid FRP rebar were broken at very irregular lengths, while the cores appeared to fracture in a regular pattern.
 - The hybrid FRP rebar specimens exhibited a large amount of slip when the peak load was reached. They continued to hold a relatively high load and continued to slip before the load was finally reduced.
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